

PhDAI 832 - LAB 1

Auditor: Nishanth Ravula

Subject: Ontological Foundations & Research Integrity Audit

Primary Source: Meert, W., De Laet, T., & De Raedt, L. (2025). "Artificial Intelligence: A Perspective from the Field." In N. A. Smuha (Ed.), *The Cambridge Handbook of the Law, Ethics and Policy of Artificial Intelligence* (pp. 17–39). Cambridge University Press.

I. Comparative Traceability Matrix

Finding	NotebookLM Generated Output	Primary Source Text (Meert et al., 2025)	Discrepancy Type
1	"System I Thinking (Fast / Instinctive): In AI, this corresponds to the subdomain of machine learning. It focuses on developing computer programs that can detect and learn statistical patterns from large datasets without being explicitly programmed."	p. 17: System I mapped to ML ("instincts, reflexes, or fast thinking"). p. 18: "without being explicitly programmed" appears in Subsection 1.2, attributed to Arthur Samuel (1959).	Page Conflation Combines claims from p. 17 and p. 18 into a single continuous assertion.
2	"This knowledge-driven approach focuses on representing information symbolically (using logic, rules, or probabilities) and performing deliberate, multi-step inference to answer new questions."	p. 17: System II mapped to reasoning and inference ("deliberate, multistep, logical, slow thinking"). p. 20: "representing information symbolically" appears three pages later in Subsection 1.3.	Page Conflation Merges structural descriptions from p. 17 and p. 20.
3	"...a recurring consensus in the field is that the capabilities of both learning and reasoning are essential to achieve intelligence."	p. 17: Text states "a reoccurring insight is that the capabilities of learning and reasoning are essential," immediately followed by discussion of ongoing debate over definitions.	Epistemic Distortion Converts a hedged "insight" into an established "consensus."
4	"Machine learning (System I) and machine reasoning (System II) developed as independent scientific tracks."	p. 17: Text notes that "most practical AI systems rely on both learning and reasoning techniques," prior to stating that each developed rather independently.	Caveat Omission Strips the contextual qualification regarding modern hybrid systems.
5	"...one of the 'grand challenges' of modern AI is achieving a truly integrated learning and reasoning mechanism (such as neuro-symbolic AI)."	p. 17: Body text notes the "grand challenge" of an integrated mechanism. The term "neuro-symbolic AI" appears exclusively in the title of a paper cited in an attached footnote.	Footnote Elevation Elevates a citation title from a footnote into main body text.

Finding 6 — Citation-Level, Upstream of the AI

Course Material	<i>"Smuha (2025) describes this as an unprecedented 'divorce' between agency and intelligence."</i>
Verified Source	Floridi (2019) opens with exactly this argument: "a divorce between agency and intelligence," later expanded into a 2023 book. Neither "divorce" nor "agency" appears anywhere in Smuha's own introductory chapter (pp. 1–13) of this Handbook.
Discrepancy Type	Author Misattribution — a course-level citation error, upstream of the AI audit entirely, tracing a claim to the wrong author.

II. Analytical Audit & Findings

This audit is a case study in the distinction between computational intelligence and moral/epistemic agency (Boddington, 2023). NotebookLM demonstrates high technical intelligence: it parsed the chapter and synthesized fluent, terminologically accurate output. It lacks agency, the capacity to be answerable for its claims. The six findings above illustrate this gap in practice.

Findings 1, 2, and 5 illustrate structural conflations rather than outright fabrications. NotebookLM extracted valid statements from separate pages and a footnote citation, stitching them together as a unified claim. While the resulting text remains technically coherent, it alters the text's original location specific structure. Elevating a footnote title into the main narrative demonstrates how automated synthesis blurs content hierarchies (Meert et al., 2025).

Findings 3 and 4 carry the primary moral weight of the audit. By converting the primary source's cautious phrase "a reoccurring insight" into "a recurring consensus," NotebookLM overstates academic agreement as an epistemic distortion rather than a stylistic choice. This is not the generic filler the lab calls "AI slop": the wording is precise and traceable to p. 17 (Meert et al., 2025), which is what makes the distortion harder to catch, not easier. Similarly, omitting the source's caveat that modern practical AI systems already combine both learning and reasoning presents an oversimplified historical binary.

Finding 6 relocates the moral weight question one level up. The assigned course material attributes the framing of AI as an unprecedented "divorce" between agency and intelligence to Smuha (2025). That specific thesis is Floridi's (2019, 2023), argued in a 2019 journal article and expanded into a 2023 book; neither "divorce" nor "agency" appears anywhere in Smuha's own introductory chapter of this Handbook. The course's citation error and NotebookLM's page conflations share the same shape, a claim traced to the wrong author.

None of NotebookLM's outputs represent traditional hallucinations; every clause traces to real text within the assigned reading. This directly demonstrates R  ther's (2025) meaningfulness gap: output that is syntactically coherent and plausible, yet distorted in ways that remain invisible without direct source verification. A passive reading of the summary would leave a researcher with a flatter, overly confident interpretation of the chapter's thesis.

III. Reflection: The Research Triad and Personal Implications

I define the research triad here as researcher, AI system, and primary source. Verification rigor exists to police the middle link the AI's mediation between reader and text but Finding 6 shows the triad's weakest point was not NotebookLM's synthesis at all, but a course citation that never touched the AI.

Running this audit changed how I read AI-generated summaries. A flat read of NotebookLM's output would have left me a marginally flatter, more confident version of the chapter's argument, and I would not have known it. An AI system cannot be held accountable for the gaps it introduces; that accountability rests on the researcher, in both the content and citation audits above.

IV. Audit Conclusion & Rule of Practice

An AI-generated summary is not a substitute for primary source analysis. Automated summaries, like unverified citations, serve as preliminary hypotheses about a text's contents and require rigorous verification against authoritative sources before scholarly or legal deployment.

References

- Boddington, P. (2023). *AI ethics: A textbook*. Springer.
- Floridi, L. (2019). What the near future of artificial intelligence could be. *Philosophy & Technology*, 32(1), 1–15. <https://doi.org/10.1007/s13347-019-00345-y>
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- Smuha, N. A. (2025). An introduction to the law, ethics, and policy of artificial intelligence. In N. A. Smuha (Ed.), *The Cambridge handbook of the law, ethics and policy of artificial intelligence* (pp. 1–13). Cambridge University Press. <https://doi.org/10.1017/9781009367783>
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